Chem 273 Discussion 5

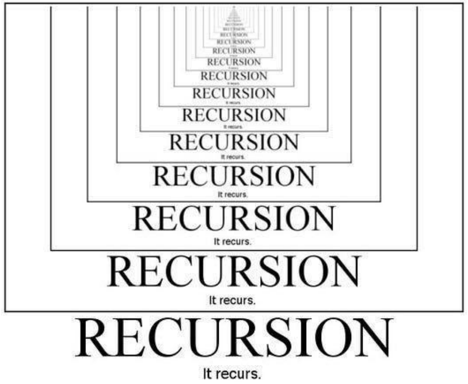
Reminders

- Lecture Exercise 05 is due Saturday, June 22

Lecture Assignment 4 Solution

Recursion

- Recursive programming is when a function calls itself
- A recursive function breaks a large problem into smaller version of the same problem
- It’s a power an alternative to loops and if statements



Recursive Functions - Example

Think about a function that counts down to 1 starting at n:

```
countdown(5)
```

5
4
3
2
1

```
countdown(3)
```

3
2
1

```
countdown(9)
```

9
8
7
6
5
4
3
2
1

Recursion

```python
def countdown(n):
    print(n)
    countdown(n-1)
```

1. countdown(3)
2. print(3)
3. countdown(2)
4. print(2)
5. countdown(1)
6. print(1)
7. countdown(0)
8. print(0)
9. countdown(-1)
10. print(-1)
11. countdown(-2)

```
3
2
1
0
-1
-2
-3
-4
-5
-6
-7
-8
-9
-10
-11
-12
-13
```

Fix the code!

```
def countdown(n):
    print(n)
    countdown(n-1)
```

Recursion

```
def countdown(n):
    if(n==1):
        print(n)
    else:
        print(n)
        countdown(n-1)
```

1. countdown(3)
2. If statement?
3. print(3)
4. countdown(2)
5. If statement?
6. print(2)
7. countdown(1)
8. If statement?
9. print(1)

Recursion

Recursive functions have two “cases”:

- The base case, the “end” result where the function is not called
- The recursive case, where the function calls itself

In the previous example what is the base case? What is the recursive case?

```
def countdown(n):
    if(n==1):
        print(n)
    else:
        print(n)
        countdown(n-1)
```

Recursion

Think about factorials:

5! = 5 * 4 * 3 * 2 * 1

What would be the base case in a recursive function for factorials?

What would the recursive case be?

Recursion

```
def factorials(n):
    # base case
    if n == 0:
        return 1

    # recursive case
    else:
        return n * factorials(n - 1)
```

1. factorials(3)
2. If statement?
3. return 3 * factorials(2)
4. factorials(2)
5. If statement?
6. return 2 * factorials(1)
7. factorials(1)
8. If statement?
9. return 1 * factorials(0)
10. If statement?
11. return 1

Bisection Method

Review: The Bisection Method is a root finding method

- It requires two initial guesses
- Repeatedly narrows down the interval where the root lies

Bisection Method

1. Choose the initial interval (values for x_left and x_right)
2. Compute the midpoint x_center = x_left + x_right / 2
3. Evaluate f(x_center), f(x_left), f(x_right)
4. Replace either x_right or x_left with x_center:

if $f(x_{center}) \cdot f(x_{left}) < 0$

- set x_{right} to x_{center}

if $f(x_{center}) \cdot f(x_{left}) > 0$

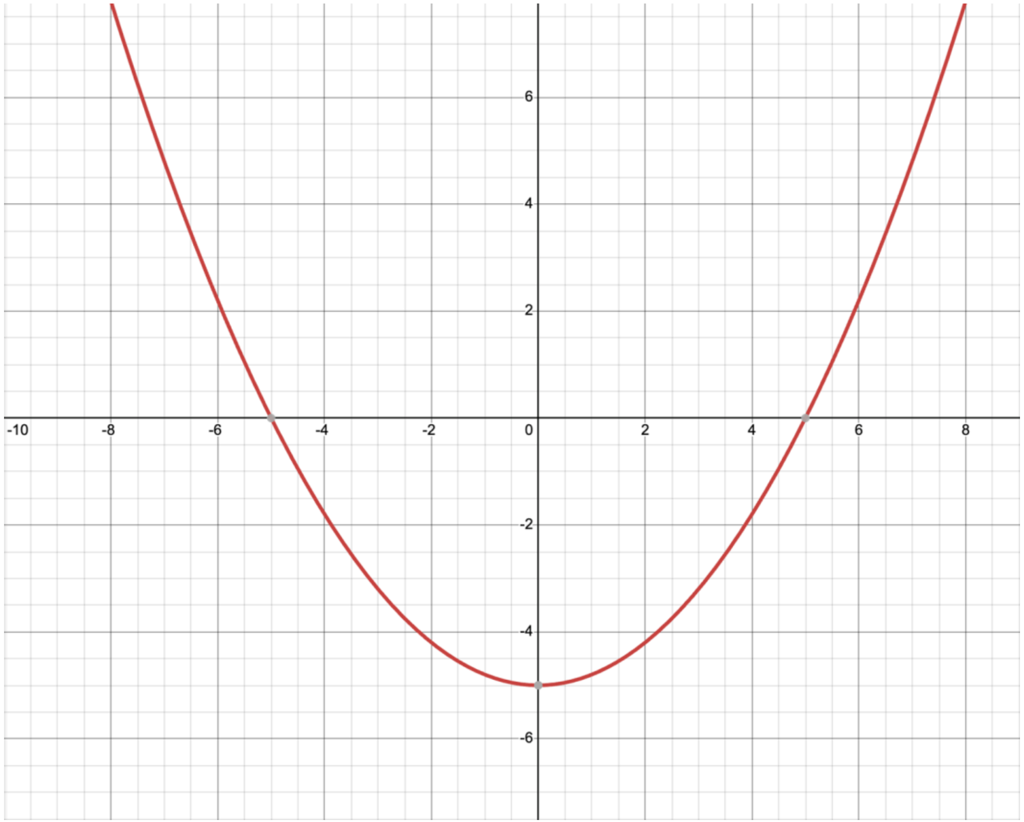
- set x_{left} to x_{center}

5. Repeat until we find the answer

Bisection Method- Example

Let’s say:

$$f(x) = \frac{1}{5}x^2 - 5$$



Bisection Method- Example

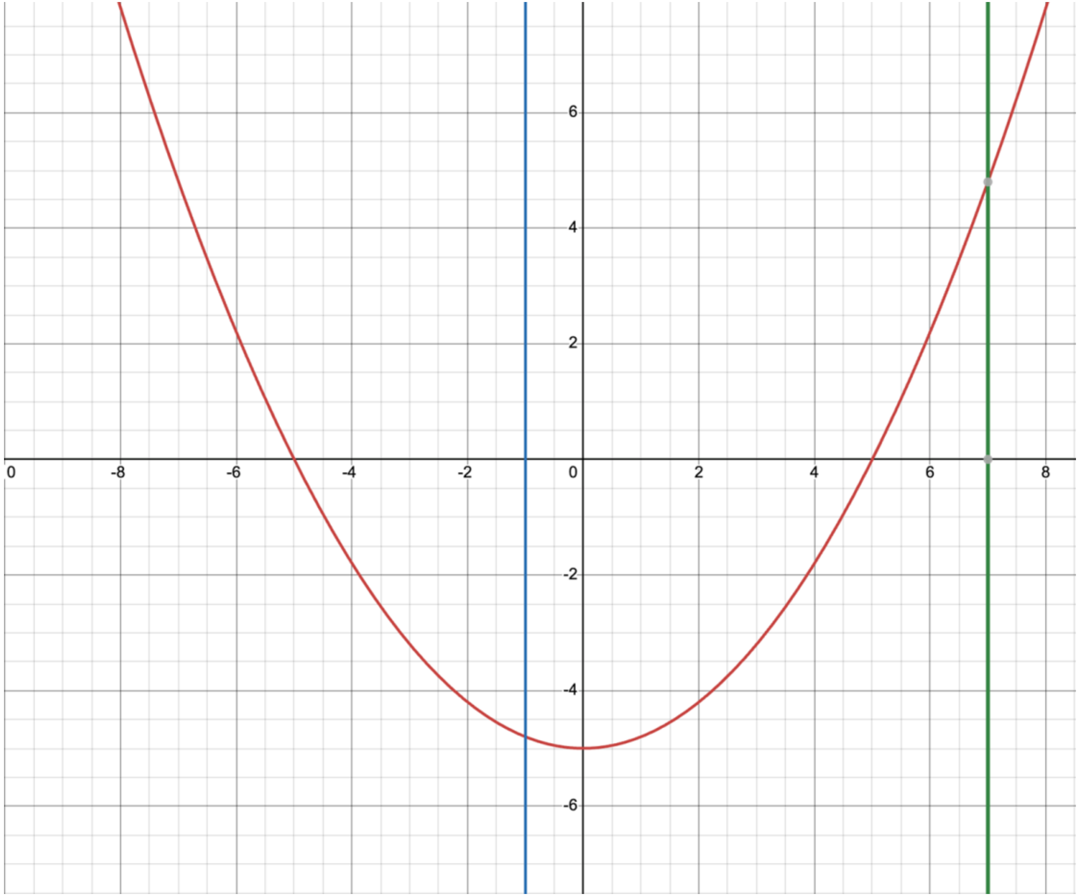
First step is to pick our left and right intervals.

x_left = -1

$$= \frac{1}{5}(-1)^2 - 5$$
$$= \frac{1}{5} - 5$$
$$= -4.8$$

x_right = 7

$$= \frac{1}{5}(7)^2 - 5$$
$$= \frac{49}{5} - 5$$
$$= 4.8$$



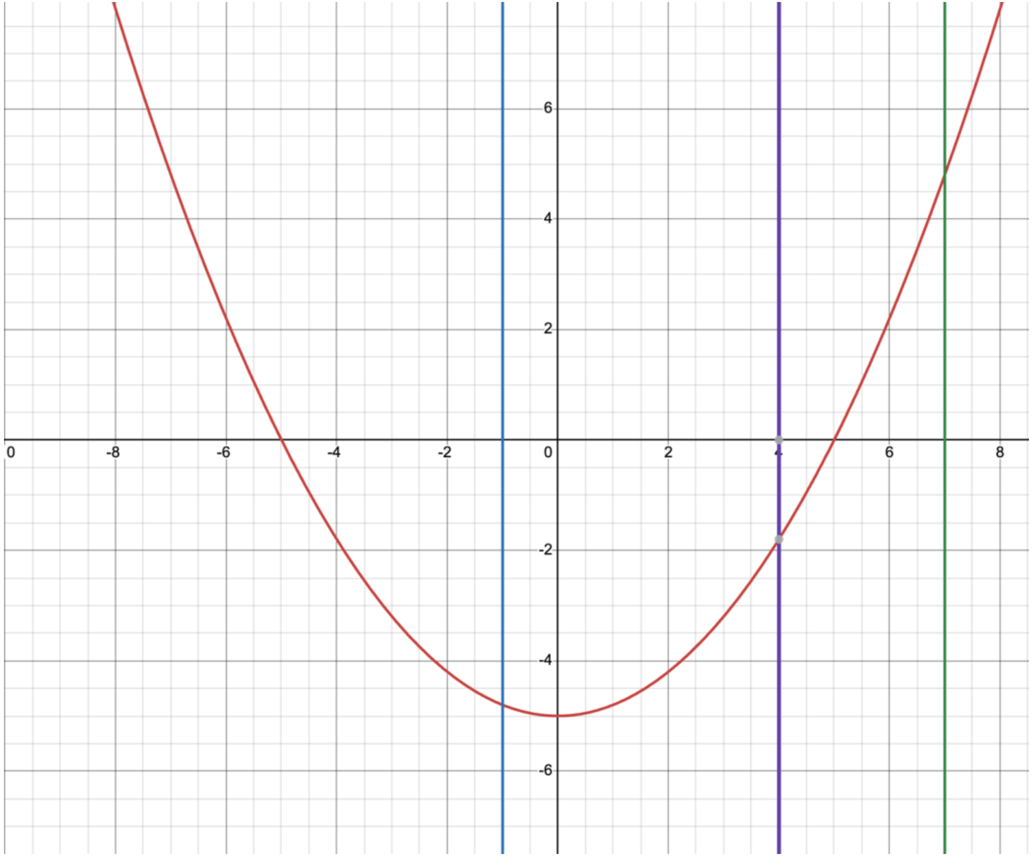
Bisection Method- Example

2. Compute the midpoint

X_left = -1

X_right = 7

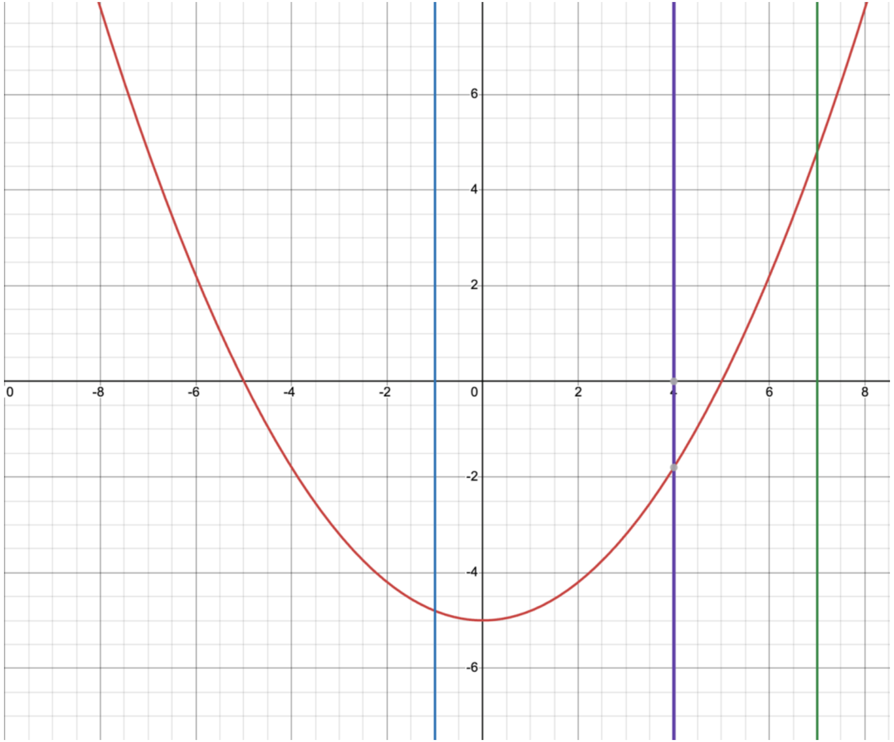
X_center = (7 + 1) / 2 = 4



Bisection Method- Example

3. Evaluate the function at x_center, x_left, x_right

$$
\begin{aligned}
f(x) &= \frac{1}{5}x^2 - 5 \\
&= \frac{1}{5}(4)^2 - 5 \\
&= \frac{16}{5} - 5 \\
&= -1.8
\end{aligned}
$$



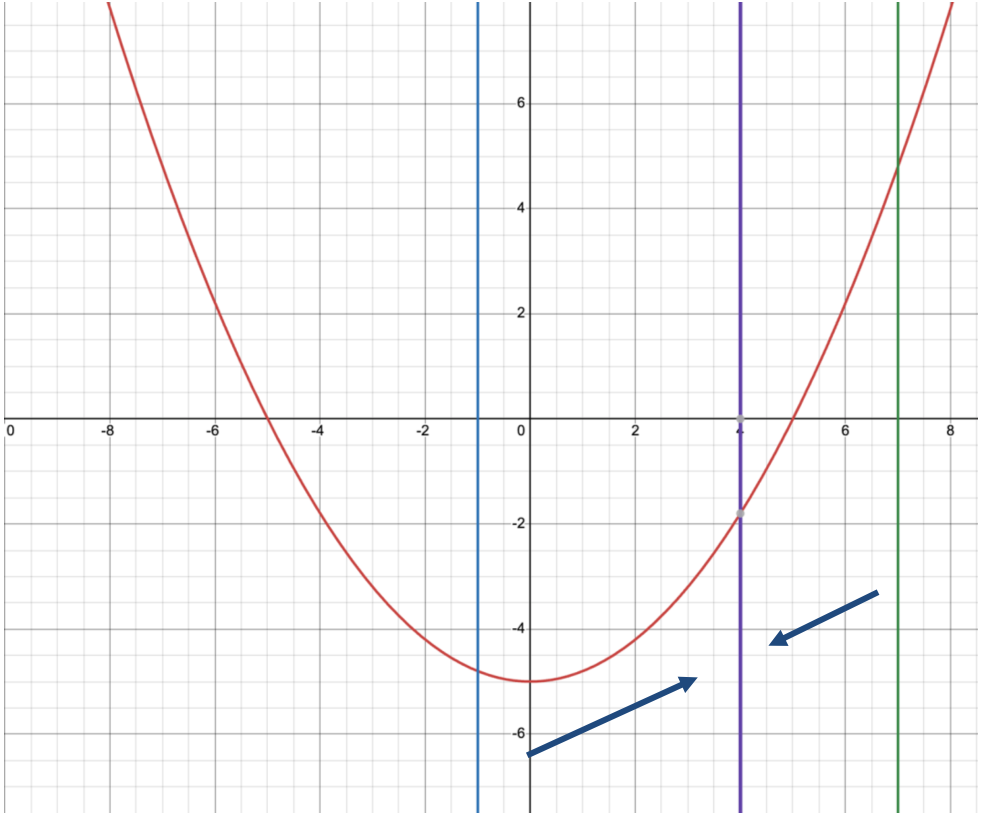
Bisection Method- Example

4. Replace either x_right or x_left with x_center.

f(x_center)	**= f(4) = -1.8**
f(x_left)	**= f(7) = 4.8**
f(x_right)	**= f(-1) = -4.8**

What happens if we replace x_left with x_center?

What happens if we replace x_right with x_center?



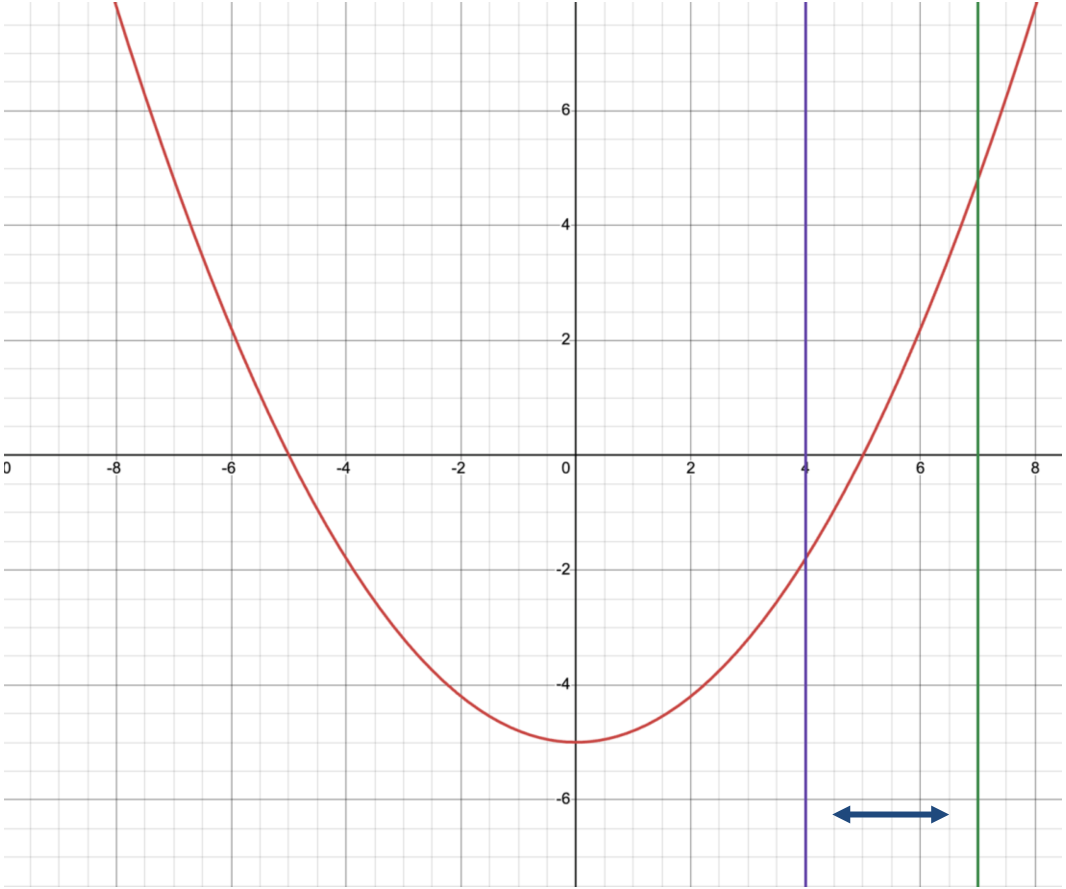
Bisection Method - Example

$$= f\left(x_{center}\right) \cdot f\left(x_{left}\right) > 0$$

-1.8 * 4.8 = -8.64 < 0

if $f(x_{center}) \cdot f(x_{left}) > 0$

- set x_{left} to x_{center}



Bisection Method- Example

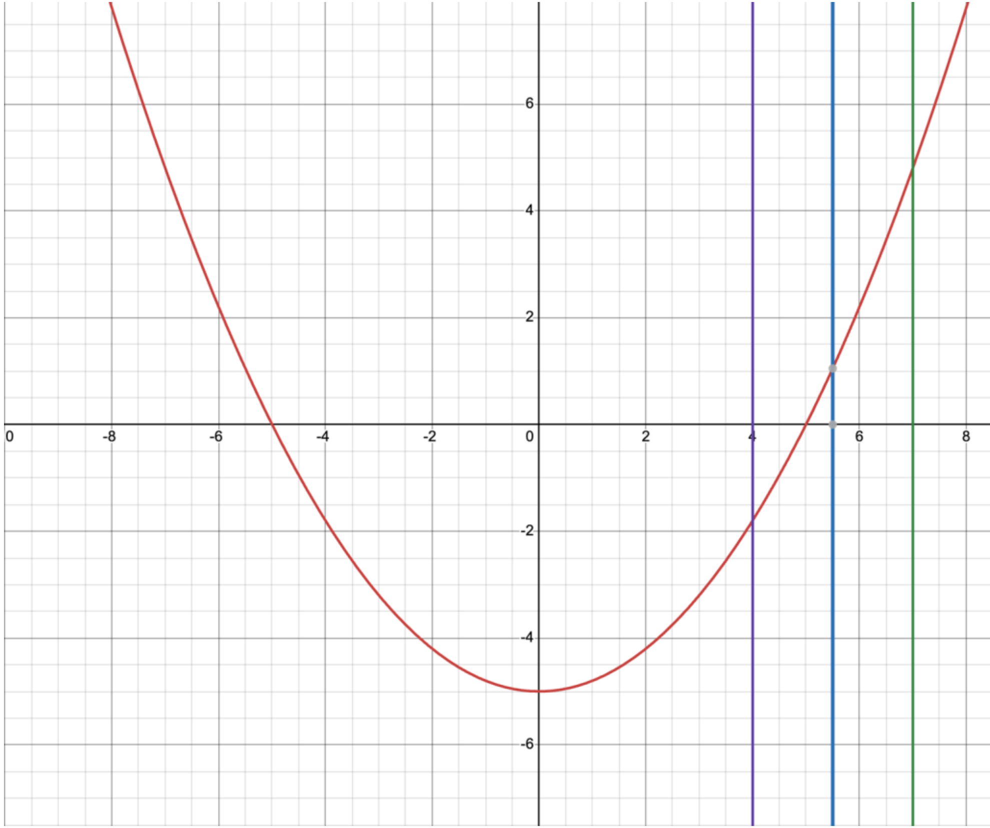
5. Repeat

x_center = (x_left + x_right) / 2
= (4 + 7) / 2 = 5.5

$$f(x) = \frac{1}{5}x^2 - 5$$

f(x_center) = f(5.5) = 1.05
f(x_left) = f(4) = -1.8
f(x_right) = f(7) = 4.8

Do we replace x_left or x_right with x_center?



Bisection Method- Example

5. Repeat

When are we done?

This process could go on forever, and never reach the true answer.

$$f\left(x_{center}\right) \cdot f\left(x_{left}\right) == 0 \qquad f\left(x_{center}\right) \cdot f\left(x_{right}\right) == 0$$

In practice we have an accuracy or tolerance, which once it's reached we are done.

Accuracy can depend on the situation, in this case, I may say my accuracy is: 0.001

$$|f(x)| < 0.001 \qquad \left|x_{right} - x_{left}\right| < 0.001$$

Function value accuracy　　　　Interval width accuracy

Lecture Exercise 05

A successful solution will:

- Use recursion, not for loops or while loops
- Return a value for the root x_center
- You can double-check your work by finding the roots of cos(x) - x and seeing if your answer is correct +/- the given accuracy

Newton's Method

To find where that tangent line crosses the x-axis, we use basic linear algebra involving the function and its first derivative.

1. Have an initial guess x_n.
2. Determine the first derivative of the function.
3. Evaluate the the function at x_n and its first derivative at x_n.
4. Solve the following equation of your estimated root.

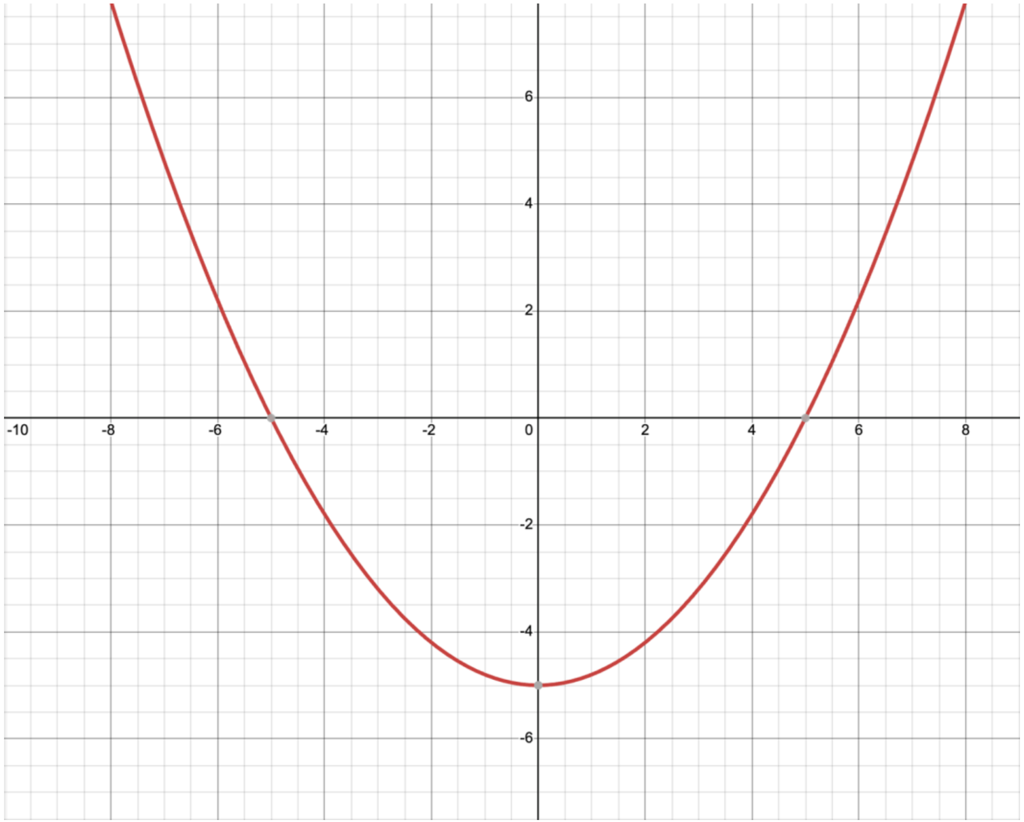
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Repeat until you are happy with your accuracy!

Newton’s Method - Example

Let’s say:

$$f(x) = \frac{1}{5}x^2 - 5$$



Newton's Method - Example

Initial guess, x_0= 3.

$$f(x) = \frac{1}{5}x^2 - 5$$

$$f'(x) = \frac{2}{5}x$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{\frac{1}{5}x_0^2 - 5}{\frac{2}{5}x_0}$$

$$x_1 = 3 - \frac{\frac{9}{5} - 5}{\frac{6}{5}}$$

$$x_1 = 3 - \frac{-3.2}{1.2}$$

$$x_1 = 3 + 2.67$$

$$x_1 = 5.67$$

Real root is x = 5.0

Newton's Method - Example

Now x_1 = 5.67

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = x_1 - \frac{\frac{1}{5}x_1^2 - 5}{\frac{2}{5}x_1}$$

$$x_2 = 5.67 - \frac{\frac{1}{5}(5.67)^2 - 5}{\frac{2}{5}(5.67)}$$

$$x_2 = 5.67 - 0.6305$$

$$x_2 = 5.04$$

Real root is x = 5.0